SAMPLE

Energy

From Kinetic to Nuclear to Dark Energy

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Introduction

Solar energy is the radiant energy from the Sun's light and heat, which can be harnessed using a range of technologies such as solar electricity, solar thermal energy (including solar water heating) and solar architecture. It is an essential source of renewable energy, and its technologies are broadly characterized as either passive solar or active solar depending on how they capture and distribute solar energy or convert it into solar power.

Active solar techniques include the use of photovoltaic systems, concentrated solar power, and solar water heating to harness the energy. Passive solar techniques include designing a building for better daylighting, selecting materials with favorable thermal mass or light-dispersing properties, and organizing spaces that naturally circulate air. In 2011, the International Energy Agency said that "the development of affordable, inexhaustible and clean solar energy technologies will have huge longer- term benefits.

It will increase countries' energy security through reliance on an indigenous, inexhaustible, and mostly import-independent resource, enhance sustainability, reduce pollution, lower the costs of mitigating global warming.... The Earth receives 174 petawatts (PW) of incoming solar radiation (insolation) at the upper atmosphere. Approximately 30% is reflected back to space while the rest, 122 PW, is absorbed by clouds, oceans and land masses.

The spectrum of solar light at the Earth's surface is mostly spread across the visible and near-infrared ranges with a small part in the near- ultraviolet. Most of the world's population live in areas with insolation levels of 150-300 watts/ m2, or 3.5-7.0 kWh/m2 per day. Solar radiation is absorbed by the Earth's land surface, oceans - which cover about 71% of the globe - and atmosphere.

Warm air containing evaporated water from the oceans rises, causing atmospheric circulation or convection.

Energy (from Ancient Greek ἐνέργεια (enérgeia) 'activity') is the quantitative property that is transferred to a body or to a physical system, recognizable in the capacity to do work and in the form of heat and light. Energy is a conserved quantity—the law of conservation of energy states that energy can be converted in form, but not created or destroyed.

The unit of measurement for energy in the International System of Units (SI) is the joule (J).

Forms of energy include the kinetic energy of a moving object, the potential energy stored by an object (for instance due to its position in a field), the

elastic energy stored in a solid object, chemical energy associated with chemical reactions, the radiant energy carried by electromagnetic radiation, the internal energy contained within a thermodynamic system, and rest energy associated with an object's rest mass. These are not mutually exclusive.

All living organisms constantly take in and release energy.

The Earth's climate and ecosystems processes are driven primarily by radiant energy from the Sun. The total energy of a system can be subdivided and classified into potential energy, kinetic energy, or combinations of the two in various ways.

Chapter 1: The Discovery of Energy

By photosynthesis, green plants convert solar energy into chemically stored energy, which produces food, wood and the biomass from which fossil fuels are derived.

The total solar energy absorbed by Earth's atmosphere, oceans and land masses is approximately 122 PW·year = 3,850,000 exajoules (EJ) per year. In 2002 (2019), this was more energy in one hour (one hour and 25 minutes) than the world used in one year. Photosynthesis captures approximately 3,000 EJ per year in biomass.

The potential solar energy that could be used by humans differs from the amount of solar energy present near the surface of the planet because factors such as geography, time variation, cloud cover, and the land available to humans limit the amount of solar energy that we can acquire.

In 2021, Carbon Tracker Initiative estimated the land area needed to generate all our energy from solar alone was 450,000 km2 — or about the same as the area of Sweden, or the area of Morocco, or the area of California (0.3% of the Earth's total land area).

Solar technologies are categorized as either passive or active depending on the way they capture, convert and distribute sunlight and enable solar energy to be harnessed at different levels around the world, mostly depending on the distance from the Equator. Although solar energy refers primarily to the use of solar radiation for practical ends, all types of renewable energy, other than geothermal power and tidal power, are derived either directly or indirectly from the Sun.

Active solar techniques use photovoltaics, concentrated solar power, solar thermal collectors, pumps, and fans to convert sunlight into useful output. Passive solar techniques include selecting materials with favorable thermal properties, designing spaces that naturally circulate air, and referencing the position of a building to the Sun. Active solar technologies increase the supply of energy and are considered supply side technologies, while passive solar technologies reduce the need for alternative resources and are generally considered demand-side technologies.

In 2000, the United Nations Development Programme, UN Department of Economic and Social Affairs, and World Energy Council published an estimate of how much of that solar potential humanity could realistically harvest each year. This took into account factors such as insolation, cloud cover, and the land that is usable by humans. It was stated that solar energy has a global

potential of 1,600 to 49,800 exajoules (4.4×1014 to 1.4×1016 kWh) per year

(see table below).

Solar thermal technologies can be used for water heating, space heating, space cooling and process heat generation. In 1878, at the Universal Exposition in Paris, Augustin Mouchot successfully demonstrated a solar steam engine but could not continue development because of cheap coal and other factors.

In 1897, Frank Shuman, a US inventor, engineer and solar energy pioneer built a small demonstration solar engine that worked by reflecting solar energy onto square boxes filled with ether, which has a lower boiling point than water and were fitted internally with black pipes which in turn powered a steam engine. In 1908 Shuman formed the Sun Power Company with the intent of building larger solar power plants. He, along with his technical advisor A.S.E.

Ackermann and British physicist Sir Charles Vernon Boys, developed an improved system using mirrors to reflect solar energy upon collector boxes, increasing heating capacity to the extent that water could now be used instead of ether. Shuman then constructed a full-scale steam engine powered by low-pressure water, enabling him to patent the entire solar engine system by 1912. Shuman built the world's first solar thermal power station in Maadi, Egypt, between 1912 and 1913.

His plant used parabolic troughs to power a 45-52 kilowatts (60-70 hp) engine that pumped more than 22,000 litres (4,800 imp gal; 5,800 US gal) of water per minute from the Nile River to adjacent cotton fields.

Although the outbreak of World War I and the discovery of cheap oil in the 1930s discouraged the advancement of solar energy, Shuman's vision, and basic design were resurrected in the 1970s with a new wave of interest in solar thermal energy.

In 1916 Shuman was quoted in the media advocating solar energy's utilization, saying: We have proved the commercial profit of sun power in the tropics and have more particularly proved that after our stores of oil and coal are exhausted the human race can receive unlimited power from the rays of the Sun. Solar hot water systems use sunlight to heat water.

In middle geographical latitudes (between 40 degrees north and 40 degrees south), 60 to 70% of the domestic hot water use, with water temperatures up to 60 °C (140 °F), can be provided by solar heating systems. The most common types of solar water heaters are evacuated tube collectors (44%) and glazed flat plate collectors (34%) generally used for domestic hot water; and

unglazed plastic collectors (21%) used mainly to heat swimming pools.

Kinetic energy is determined by the movement of an object - or the composite motion of the object's components - while potential energy reflects the potential of an object to have motion, generally being based upon the object's position within a field or what is stored within the field itself.

While these two categories are sufficient to describe all forms of energy, it is often convenient to refer to particular combinations of potential and kinetic energy as its own form.

For example, the sum of translational and rotational kinetic and potential energy within a system is referred to as mechanical energy, whereas nuclear energy refers to the combined potentials within an atomic nucleus from either the nuclear force or the weak force, among other examples. The word energy derives from the Ancient Greek: ἐνέργεια, romanized: energeia, lit. 'activity, operation', which possibly appears for the first time in the work of Aristotle in the 4th century BC.

In contrast to the modern definition, energeia was a qualitative philosophical concept, broad enough to include ideas such as happiness and pleasure. In the late 17th century, Gottfried Leibniz proposed the idea of the Latin: vis viva, or living force, which defined as the product of the mass of an object and its velocity squared; he believed that total vis viva was conserved.

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ABOUT THE AUTHOR

Aira Astralis is a science writer specializing in astronomy and space exploration. Her work spans the full breadth of humanity's relationship with the cosmos -- from the earliest naked-eye observations to the engineering marvels of modern space missions.

Her books are built on a foundation of primary sources: peer-reviewed research, mission archives, and direct engagement with the scientific community. She writes for readers who want to understand not just what we know about the universe, but how we came to know it.

She believes that the most important frontier is not out there but in the gap between what scientists discover and what the public understands. Her books exist to close that gap.